1. PROBLEM STATEMENT

Urban residents and visitors lack access to comprehensive, real-time safety intelligence when navigating cities. Traditional crime maps and dashboards typically plot reported incidents after they occur and rarely provide predictive insights that combine incident severity, temporal dynamics, and location context. This limits the ability of citizens, planners, and responders to proactively assess risk for a particular time and place.

The proposed project, **Aegis**, aims to provide a data-driven predictive safety score for urban locations by combining historical crime records with an evidence-based severity weighting (from the Bureau of Justice Statistics), temporal features, and location descriptors. The output is a continuous "unsafety" score (a higher value means higher weighted risk) that can be used by visualization layers (heatmaps, choropleths) or downstream services (APIs for route-safety queries).

The absence of a weighted, ML-driven system limits predictive capability and does not exploit authoritative severity information (e.g., BJS) to prioritize incidents when estimating location risk.

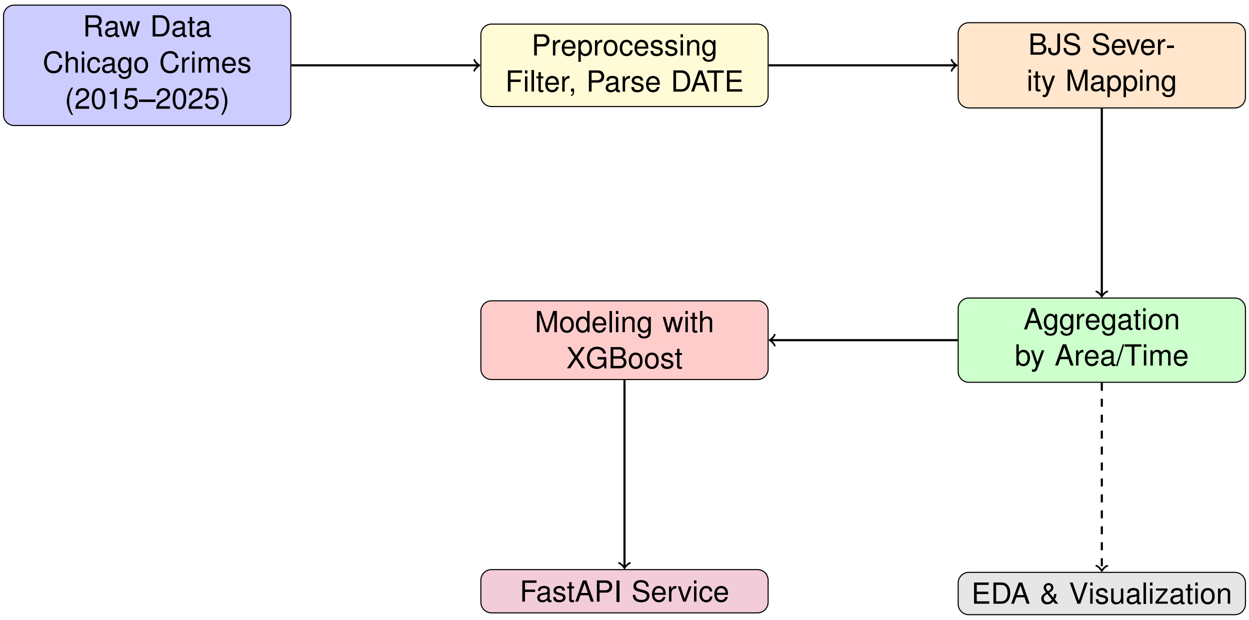
2. OBJECTIVES

1. Build a reproducible machine learning pipeline that predicts a continuous
2. weighted_crime_score per (community area, month, day_of_week, hour, location_description).
3. Use an authoritative severity mapping (BJS National Survey of Crime Severity) to assign weights to crime records and derive a defensible target variable.
4. Limit the training data to the period 2015–2025 to reflect contemporary patterns and keep the dataset manageable.
5. Provide exploratory analyses (time series, hourly/daily patterns, choropleth maps) to validate weighting and highlight hotspots.
6. Train a Gradient Boosting regressor (e.g., XGBoost) to predict the weighted score using community, temporal and location features.
7. Serve predictions via a lightweight API (FastAPI) for integration with visualization frontends (heatmap/choropleth).

3. INTRODUCTION

Aegis reframes urban safety assessment from static incident plotting into a predictive, severity-weighted regression problem. The solution combines the City of Chicago "Crimes - 2001 to Present" dataset with a severity mapping derived from the Bureau of Justice Statistics (BJS) National Survey of Crime Severity. The dataset is filtered to 2015–2025 to maintain relevance.

Aegis' outputs are continuous weighted scores aggregated by meaningful spatio-temporal groupings (community area, month, day, hour, location description). This design supports multiple applications: visualization (heatmaps, choropleths), temporal alerts for planners, and API-based integration for routing or mobile clients.



4. METHODOLOGY

4.1. 1.0 Data Acquisition and Preparation

- **Data Source:** City of Chicago "Crimes - 2001 to Present" dataset. We restrict to 2015–2025 to reflect modern patterns and to keep processing tractable.
- **Timeframe Filtering:** Filter rows where DATE falls within 2015-01-01 to 2025-12-31.
- **Feature Selection:** Retain columns: `ID`, `DATE`, `PRIMARY_TYPE`, `DESCRIPTION`, `LOCATION_DESCRIPTION`, `COMMUNITY_AREA`, and coordinates if needed for mapping.
- **Datetime Processing:** Parse `DATE` into `YEAR`, `MONTH`, `DAY_OF_WEEK`, and `HOUR`.
- **Data Cleaning:** Remove rows lacking community area or location description; handle missing or malformed timestamps; standardize categorical names.

4.2. Weighted Score Calculation (Using BJS Data)

- Use the BJS National Survey of Crime Severity (NSCS) as the authoritative basis to assign severity weights by crime type and description.
- Create a `bjs_severity_score` column for each crime record by mapping (`PRIMARY_TYPE`, `DESCRIPTION`) $\rightarrow$ severity weight according to the extracted BJS mapping table.
- If an exact mapping is not available for a fine-grained description, map to the closest `PRIMARY_TYPE` severity and document assumptions.
- **Example:** Map "THEFT - OVER $500" to a higher theft weight from NSCS; map "THEFT - RETAIL THEFT" to a lower theft severity.

4.3. Target Variable Creation

1. Group data by `COMMUNITY_AREA`, `MONTH`, `DAY_OF_WEEK`, `HOUR`, and `LOCATION_DESCRIPTION`.
2. Sum `bjs_severity_score` within each group to produce the aggregated weighted_crime_score. This continuous numeric sum is the target for modeling (higher $\Rightarrow$ more "unsafety").
3. Optionally, compute counts and normalized rates (per-capita or per-1000 population) if population denominators are available for normalization analyses—keep the primary target as the aggregated weighted score.

4.4. 2.0 Exploratory Data Analysis (EDA)

EDA validates assumptions and provides insights that inform model features and transformations:

- **Crime Trends Over Time:** Time series of aggregated weighted score to identify seasonal or multi-year trends.

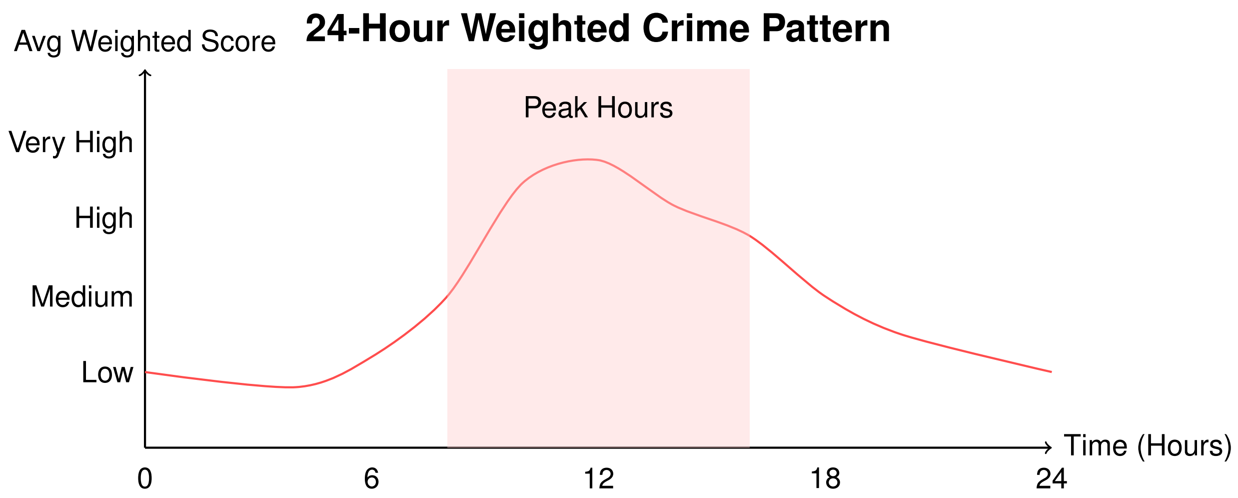
- **Hourly and Daily Patterns:** Bar charts of average weighted score by hour and by day-of-week to highlight peak-risk windows.
- **Geographic Hotspots:** Choropleth of community areas colored by mean weighted score to find persistent hotspots.
- **Location Descriptors:** Analyze distribution of `LOCATION_DESCRIPTION` (e.g., STREET, SIDEWALK, RESIDENCE) and their average severity contributions.
- **Outlier Detection:** Identify unusual spikes (reporting anomalies) and apply temporal smoothing or trimming rules as needed.

4.5. 3.0 Model Development and Training

- **Problem Framing:** Regression — predict continuous `weighted_crime_score`.
- **Feature Set:** One-hot or categorical-encoded `COMMUNITY_AREA`, `MONTH`, `DAY_OF_WEEK`, `HOUR`, and `LOCATION_DESCRIPTION`. Consider interaction features (e.g., community_area $\times$ hour) if data supports it.
- **Train/Test Split:** Temporal-aware split is recommended (e.g., train on 2015–2023, test on 2024–2025) to evaluate forward performance, or a randomized 80/20 split if time leakage is controlled.
- **Model Choice:** XGBoost (Gradient Boosting Regressor) — effective on structured tabular data and handles heterogeneous categorical encodings when preprocessed properly.
- **Training:** Use early stopping with a validation set and tune hyperparameters (learning rate, max depth, number of estimators, subsample).
- **Feature Importance:** Extract feature importance to explain model behavior (which temporal or location features drive predictions).

4.6. 4.0 Evaluation and Deployment

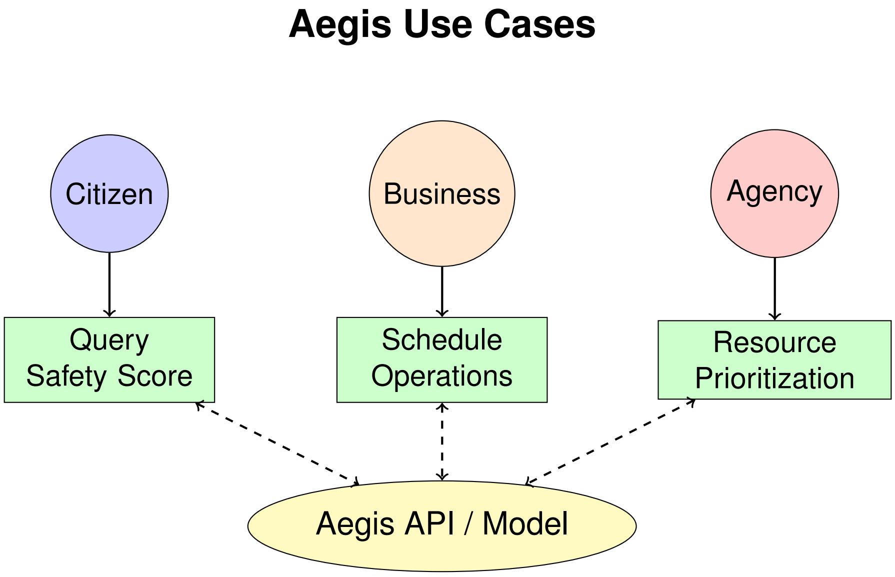
- **Evaluation Metrics:** Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) on the test set. MAE gives interpretable average error in weighted-score units.
- **Calibration:** Check predicted distributions versus actuals; if required, apply post-processing (e.g., isotonic regression) to improve calibration.
- **API Development:** Implement a FastAPI service that accepts `community_area`, `month`, `day_of_week`, `hour`, and `location_description` and returns the predicted `weighted_crime_score` and derived safety level (e.g., quantile / color).
- **Deployment:** Package model artifact (e.g., XGBoost binary or ONNX) and deploy with the FastAPI app on a cloud service (AWS/GCP/Azure). Use Redis caching for frequently requested queries and set up simple monitoring (error rates, response latency).


24-Hour Weighted Crime Pattern
Avg Weighted Score
Peak Hours
Very High
High
Medium
Low
Time (Hours)
0
6
12
18
24

5. IMPORTANCE AND APPLICATIONS

5.1. Immediate Benefits

- **Personal Safety:** Allows individuals to query a predictive weighted score for a time and location, enabling informed decisions (e.g., choosing when to travel).
- **Tourism and Commerce:** Helps visitors and businesses understand temporal risk patterns in districts.
- **Data-Driven Policing and Planning:** Supports agencies and planners with severity-weighted heatmaps to prioritize resources.
- **Analytic Insights:** Provides statisticians and city analysts with defensible, severity-weighted trend analyses.



5.2. Long-term Impact

- Improve urban planning and allocation of safety resources via a severity-weighted lens.
- Provide a standard, defensible approach to convert crime records into actionable continuous measures.
- Support cross-city comparisons if the severity mapping is standardized and data harmonized.

6. TOOLS AND TECHNOLOGIES

Our technology stack is selected for reproducibility, interpretability, and operational readiness.

6.1. Frontend Development

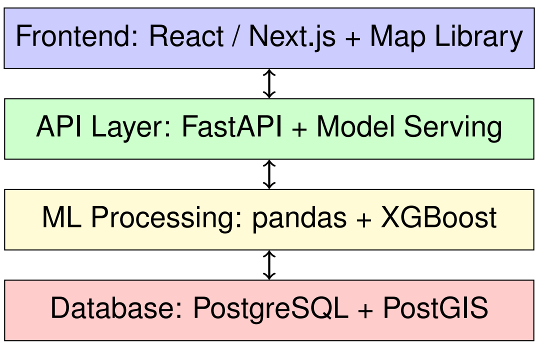
- **Visualization:** Leaflet.js or Mapbox for interactive choropleths and heatmaps (frontend reads Aegis API).
- **Web Platform:** Next.js or plain React for dashboards and embedding maps.

6.2. Backend Infrastructure

- **API Service:** FastAPI (Python) for low-latency prediction endpoints.
- **Database:** PostgreSQL + PostGIS for spatial joins and choropleth generation; use raw SQL for aggregation workflows.
- **Caching / Queue:** Redis for cached queries; optional Celery for batch aggregation jobs.

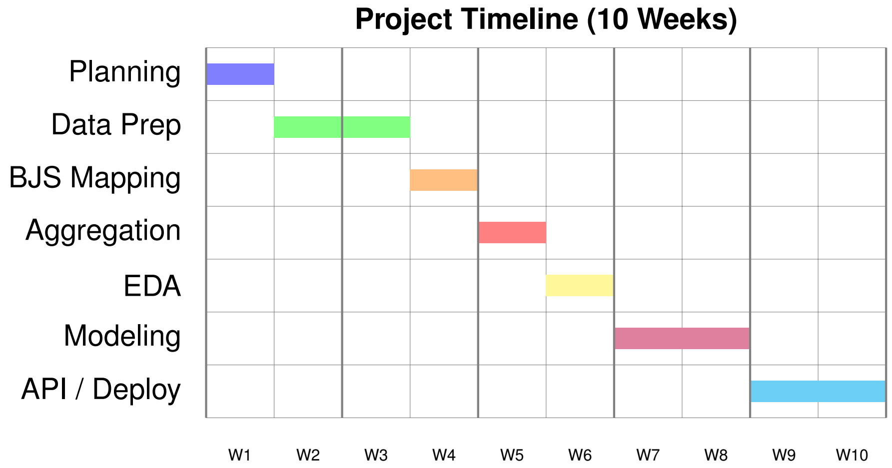
6.3. Data Processing and ML

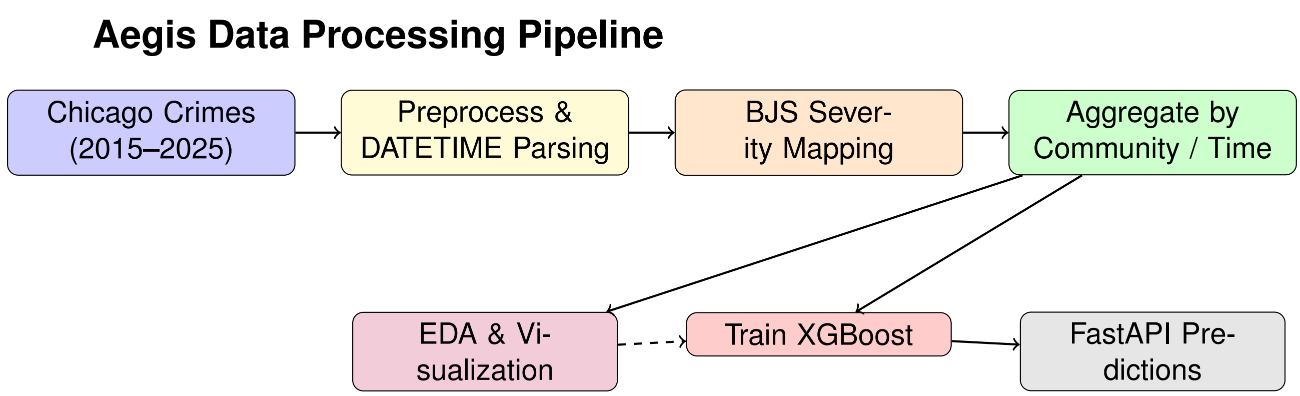
- **Data Wrangling:** pandas, NumPy for ETL and aggregation.
- **Modeling:** XGBoost (or LightGBM) as the primary regressor.
- **Deployment:** Model saved as binary (XGBoost booster) or exported to ONNX, served by FastAPI.



7. TIME SCHEDULE

Phase	Activities
Week 1	Planning, requirements analysis, dataset access
Week 2	Data filtering (2015–2025), feature selection, datetime parsing
Week 3	Build BJS severity mapping and compute `bjs_severity_score`
Week 4	Aggregation pipeline and target variable creation
Week 5	Exploratory Data Analysis (time series, hourly/daily patterns, choropleths)
Week 6-7	Model development (XGBoost), hyperparameter tuning, validation
Week 8	Feature importance analysis and calibration
Week 9	FastAPI development and model serialization
Week 10	Deployment, caching setup, monitoring, final evaluation/reporting





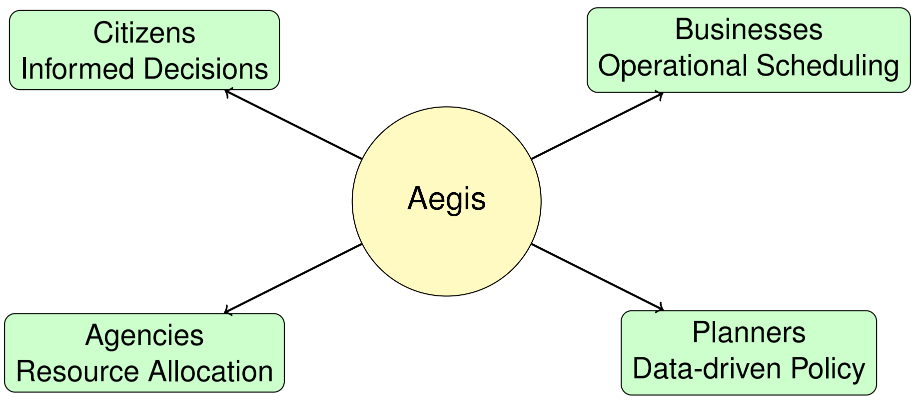
8. FUTURE SCOPE

- **Multi-city Deployment:** Extend Aegis by harmonizing severity mappings and deploying on other city datasets.
- **Advanced Models:** Explore deep learning/time-series models (Temporal Fusion Transformer) if sequence-based predictions are needed.
- **Real-time Ingestion:** Integrate streaming ingestion for near real-time aggregated scores (e.g., using Kafka).
- **Personalized Alerts:** Allow users or services to subscribe to periodic risk summaries for frequented community areas.
- **Normalization Layers:** Add population or footfall normalization to convert absolute weighted sums into per-capita or exposure-adjusted rates.

9. CONCLUSION

Aegis provides a rigorous, severity-weighted, machine-learning approach to predict a continuous measure of location-level "unsafety." By anchoring weights to the BJS NSCS and modeling aggregated scores across meaningful spatio-temporal slices, Aegis produces defensible predictions useful for visualization, policy prioritization, and integration via API services. The planned pipeline (preprocessing, BJS-weight mapping, aggregation, EDA, XGBoost modeling, and FastAPI serving) ensures reproducibility and operational readiness.

Aegis Impact Summary



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[4] Sebastian Ramirez. FastAPI documentation and best practices. https://fastapi.tiangolo.com

[5] Lundberg, S. M., & Lee, S.-I. (2017). "A Unified Approach to Interpreting Model Predictions." In *Advances in Neural Information Processing Systems*.